

17/11/2014(E)

K. J. Somaiya College of Engineering, Mumbai-77

(Autonomous College Affiliated to University of Mumbai)

Semester: July - November 2014

Max. Marks: 100

Duration: 3hr.

End Semester Examination

Class: SE Comp A&B

Semester: III

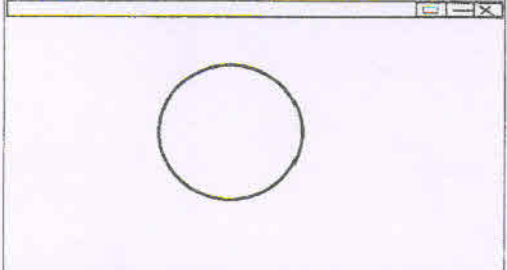
Name of the Course: Object Oriented Programming Methodology

Branch: Computer

Course Code: UCEC302

NOTE: 1) Draw neat & Clean Diagram.

2) All Questions are compulsory.

Questions No.	Questions	Max. Marks
Q1.	a) Construct the class diagram for cellular phone.	10
	b) P.T.O	P.T.O
Q 2	Write short notes on: (Any Four) i) Encapsulation ii) Inheritance iii) Object iv) Abstraction v) Polymorphism vi) Metaclass vii) JVM	20
Q 3	a) Explain the concept with example Branching and looping.	10
	b) What is the concept of Method Overloading and Method overriding. in brief with suitable example?	10
Q 4	a) i) State a note on Abstract class and abstract Inheritance. ii) Write a program for String Manipulation.	10
	b) Explain the following with suitable one Real-Life examples for each Aggregation, Generalization, Association, Role, and Qualifier. OR Write a program for Multithreading concept using basic JAVA language.	10
Q 5	a) Write note on: (Any Two) i) Wrapper class ii) Array List iii) Linked List. OR Explain the Exception handling with Example.	10
	b) Write a Applet program for following output 	10

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Q1 b] Implement the above class diagram of Q1.(a)

(10)

OR

Write a Program To implement following matrix operation using interface

```
interface matrix {  
    int m, n;  
    add ();  
    multiply ();  
}
```

(10)